

EMRE OZAN ORAL

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SUMMARY

Computer Science graduate with a strong foundation in programming (C++, data structures, algorithms) and hands-on experience building AI and LLM-based systems, including retrieval-augmented generation pipelines, multi-agent orchestration, and deep learning models. Comfortable moving between low-level implementation and applied AI/ML engineering.

WORK EXPERIENCE

Promake AI — AI Developer (Part-Time)

Sep 2025 – Dec 2025

- Designed and implemented a multi-agent LLM system for automated website generation using LangGraph
- Built agent orchestration pipelines with tool integration and dynamic routing
- Developed scalable prompt workflows for structured website creation

SESTEK — AI R&D Intern

Jun 2025 – Aug 2025

- Built and evaluated LLM-based multi-agent systems using LangGraph and DeepEval
- Implemented agent-based pipelines with open-source LLM tools (Ollama, Groq)
- Conducted evaluation experiments to measure agent performance and reliability

PROJECTS

InfoGuide — RAG-Based AI Information Retrieval System (Graduation Project)

- Designed and implemented an end-to-end Retrieval-Augmented Generation (RAG) system for financial knowledge retrieval
- Built a full pipeline including document ingestion, preprocessing, embedding generation, and vector indexing (FAISS / ANN)
- Integrated LLM-based answer generation with retrieved context to ensure grounded and traceable outputs
- Developed evaluation pipeline using retrieval metrics (Recall@k) and qualitative assessment of generated responses
- Implemented scalable workload execution and benchmarking using HPC infrastructure (SLURM, GPU parallelization)

Image Super-Resolution with ESRGAN (EE417 — Deep Learning)

- Implemented an ESRGAN-based super-resolution model in PyTorch, using a generator built on Residual-in-Residual Dense Blocks (RRDB) and an adversarial discriminator
- Trained the model in two phases — a pixel-loss pretraining stage followed by GAN fine-tuning — on the DIV2K dataset at 4x upscaling
- Optimized training with mixed-precision computation and gradient accumulation to fit larger effective batch sizes on limited GPU memory
- Built an inference pipeline supporting single-image, batch, and tiled processing for large images, with PSNR/SSIM evaluation against bicubic baselines

AI Website Builder (Multi-Agent System)

- Built an LLM-based system that generates full websites using multiple agents
- Designed supervisor-agent architecture for task routing and execution
- Integrated tools and dynamic prompts for structured outputs

Hospital Appointment Application (Android)

- Developed a mobile application for hospital appointment scheduling
- Implemented user interaction, scheduling logic, and data handling
- Applied OOP principles and teamwork in development process

EDUCATION

Sabancı University — Bachelor of Science in Computer Science

Sep 2021 – Jun 2026

- Full Scholarship
- Core coursework: Object-Oriented Programming, Data Structures & Algorithms, Systems Programming (C++), Machine Learning & AI

Sırrı Yırcalı Anatolian High School

- One year of extra English – German classes

TECHNICAL SKILLS

Languages: C++, Python, C#, Verilog, Java

AI / ML: PyTorch, LangChain, LangGraph, DeepEval, Scikit-learn

Tools & Technologies: Git, Docker, Ollama, OpenAI API

ACTIVITIES

- Computer Science Society – Sabancı University
- Game Development Club – Sabancı University

INTERESTS

Artificial Intelligence, Robotics, Cars & Automotive Tech, Game Development, Music, Swimming